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PROFESSIONAL SUMMARY:

- Over 8+ years of experience in **Data Engineering, ETL workflows, and Data Visualization**, I have developed a deep expertise in handling large-scale data systems, optimizing workflows, and transforming raw data into actionable insights across multiple industries, including **finance, energy, and healthcare**.
- Experienced in designing scalable ETL processes using **Apache Spark, PySpark, and SQL** to deliver robust data solutions.
- Worked on developing and enhancing data ingestion and processing frameworks using **Azure Data Factory, Databricks, AWS, and Google Cloud Platform (GCP)**.
- Proficient in designing and managing enterprise data pipelines and models using **Azure Data Factory (ADF), Microsoft MDS, and Snowflake**, with expertise in **SQL, stored procedures, and data warehousing concepts**, ensuring smooth integration across cloud platforms.
- Experienced in using **Power BI and Tableau** to create insightful dashboards and visualizations that improve executive reporting and support data-driven decision-making.
- Experienced in machine learning, big data technologies, and development using **R and Python**, along with proficiency in **Linux, SQL, and Git/GitHub**.
- Extensive knowledge of the **Hadoop** ecosystem components, combined with hands-on experience in processing and analyzing large-scale datasets using **Spark, Hive, and YARN**.
- Created efficient data models for optimized storage and retrieval using **Apache Hive, Apache Parquet, and Snowflake**.
- I have expertise in Azure Data Factory (ADF) Mapping Data Flow components like Join, Lookup, Filter, and Sort. These tools simplify data processing and transformations.
- With this knowledge, I create code-free ADF solutions that work seamlessly with SSIS, improving data workflows and efficiency.
- Successfully migrated multiple **SSIS** packages to the cloud from on-premises using a "**Lift and Shift**" strategy, with minimal configuration changes as required.
- Developed **ETL pipelines** to enable smooth data integration between **OLAP and OLTP** environments, ensuring timely and precise data flow for analytics and reporting.
- Proficient in creating automation regression scripts using Python to validate ETL processes across various databases, including **Oracle, SQL Server, Hive, and MongoDB**.
- Worked on programming languages like Python, **Scala, and Java**, with practical experience in containerization technologies such as **Docker and Kubernetes** to facilitate cloud deployments.
- Strong knowledge of relational and **NoSQL** databases, including **PostgreSQL, Oracle, and MongoDB**, along with Infrastructure as Code (IaC) experience using **Terraform**.
- Developed and deployed reporting solutions using **Oracle SQL and PL/SQL** to extract valuable insights from **JSON and XML data**, seamlessly integrating with **Tableau and other BI tools** to create **interactive dashboards** and support data-driven decision-making.
- Implemented **data security measures and access controls** for **Oracle-based Jasper Reports**, ensuring regulatory **compliance** and secure data access.
- Collaborated with stakeholders throughout the software development lifecycle (SDLC), using agile methods to gather requirements and develop effective solutions.
- My focus was on optimizing resources, reducing costs, and ensuring compliance with industry regulations to improve efficiency and meet business needs.

EDUCATIONAL EXPERIENCE:

- Master's in Business Analytics at the Governors State University, Illinois, United States. (2023)
- Bachelor's in Business Administration, India. (2017)

TECHNICAL SKILLS:

Programming Languages	SQL, PL/SQL, Python (including Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch)
Reporting/BI Tools	MS Office, Tableau, Tableau Server and Reader, Power BI, SSRS
BigData Tools	Apache Spark, Hadoop, Azure Data Lake Storage, Azure Data Factory, Azure Databricks
Data Warehousing and Business Intelligence	Data modeling in dimensional structures - Star/Snowflake schema, ETL, Change Data Capture CDC, OLAP, OLTP mostly using PL/SQL as manual ETL.
Database Systems	MySQL, Microsoft SQL Server, PostgreSQL, MongoDB, DynamoDB
Cloud Platforms	Amazon Web Services (AWS), MS Azure, Google Cloud Platform (GCP)
Frameworks	Apache Spark, Apache Kafka, Apache Airflow, Hadoop
Web Technologies	Amazon Redshift, Snowflake, BigQuery
Web Services	Amazon S3, AWS Glue, Amazon Kinesis, AWS Lambda
Web/Application Servers	Apache HTTP Server, Apache Tomcat
Operating Systems	Windows, MacOS, Unix, Linux
Version Control	Git, GitHub
Methodologies	Data Modelling, Data Pipelines, ETL processes, Data Warehousing

PROFESSIONAL EXPERIENCE:

Client: Fidelity Investments, Boston, MA (Remote)

Jan 2024 - Present

Role: Sr. Data Engineer

Responsibilities:

- Designed and built a financial data lake for the **Implementing of a Financial Data Lake** design, utilizing **Azure Data Lake** and **Azure Synapse Analytics** to store and analyze large-scale transactional data in real-time, streamlining financial data processing workflows.
- Worked on designing and managing enterprise data pipelines and models using **Azure Data Factory (ADF)**, **Microsoft MDS**, and **Snowflake**, with expertise in **SQL**, **stored procedures**, and **data warehousing concepts**, ensuring smooth integration across cloud platforms.
- Developed a comprehensive data validation framework to ensure data integrity across **Azure SQL**, **Snowflake**, and **Azure Data Lake**, incorporating **stored procedures and views for transformation logic**.
- Worked on the creation of **Cloud-Based Data Solutions** for **Investment Infrastructure** design, modernizing the company's investment data infrastructure by migrating on-premises systems to **AWS** to enhance data processing, scalability, and operational efficiency.
- Built a real-time fraud detection model using **Python**, **Spark Streaming**, and **Databricks** to process data and generate alerts for fraud detection, enhancing risk monitoring.
- The objective was to simplify **data integration** from multiple sources, enabling real-time analysis and reporting of investment portfolios and other financial services.
- Created and implemented complex data solutions using **AWS services** like **EMR**, **S3** and **Glue** ensuring efficient data processing and integration.
- Worked on developing Kafka producers to stream data from external **REST APIs** into Kafka topics.
- Combined the GraphQL API with existing REST services, enabling smooth data retrieval from multiple sources.

- Designed and managed scalable, cloud-based data pipelines using **AWS** services like Glue, **EKS**, **Redshift**, **PySpark**, **Lambda**, and **Snowflake**.
- Developed data processing workflows with **Spark**, and **AWS EMR** to manage large datasets and enhance performance in distributed computing environments.
- Created and deployed data pipelines with **PySpark** and Airflow to streamline data processing and automate workflows effectively.
- Leveraged **S3**, DynamoDB, and **RDS** for optimized data management and storage, ensuring the security and scalability of solutions.
- Created and implemented **comprehensive data quality checks** using custom SQL scripts, Python frameworks, and automation tools to maintain high data integrity and consistency.
- Collaborated with cross-functional teams to collect and analyze business requirements, transforming them into scalable technical solutions that adhere to financial regulatory standards.
- Created and managed data pipelines using **AWS Glue** and Apache Airflow to orchestrate ETL processes, ensuring data accuracy and timely availability for stakeholders.
- Utilized relational databases **MySQL** to efficiently store and manage structured data.
- Applied **Infrastructure as Code (IaC)** principles using tools like **Terraform** to automate the deployment and management of **AWS** resources, ensuring consistency and reliability in the cloud environment.
- Implemented data pipelines with Docker and Kubernetes, ensuring scalability, reliability, and fault tolerance within a distributed cloud infrastructure.
- Reduced cloud infrastructure costs by implementing auto-scaling policies, resulting in a 25% decrease in AWS operational expenses while ensuring high system availability and reliability.

Environment: AWS, Azure, Apache Kafka, Spark, Hadoop, PySpark, Terraform, Docker, SQL, Python, Power BI.

Client: United Health, Chicago, IL (Hybrid)

Jan 2023 - Dec 2023

Role: Data Engineer

Responsibilities:

- Designed and optimized financial data pipelines in a **Large-Scale Data Transformation and Cloud Migration** initiative, improving reporting efficiency and enabling real-time analytics by transitioning from on-premises SQL databases to a cloud-based infrastructure.
- Designed and managed enterprise-level data pipelines and models utilizing Azure Data Factory (ADF), Microsoft MDS, and Snowflake. Demonstrated strong expertise in SQL, stored procedures, and data warehousing principles to ensure seamless integration across cloud environments.
- Built a robust data validation framework to maintain data integrity across Azure SQL, Snowflake, and Azure Data Lake, using stored procedures and views to implement transformation logic.
- Worked on the migration of on-premises **SQL databases** to a cloud-based infrastructure with **Snowflake** and **Azure Data Factory**, ensuring smooth data integration and optimized performance.
- Developed and enhanced **ETL** pipelines using Azure Data Factory and **PySpark** for incremental data ingestion, boosting processing efficiency and minimizing cloud expenses.
- Leveraged RESTful APIs to provide access to key functionalities, enabling external applications to retrieve portfolio and transaction data.
- Developed **Snowflake-based data integration using Azure Data Factory**, implementing scalable solutions aligned with modern cloud data warehousing best practices.
- Implemented delta ingestion, data quality frameworks, and table-level security with **Databricks** and Azure Data Factory, enhancing incremental data processing efficiency.
- Built **Power BI** dashboards with real-time data refresh from **Azure Data Lake**, improving data visualization and executive reporting, while working closely with cross-functional teams to address technical challenges and ensure compliance with data policies.

- Successfully migrated on-premises SQL databases to **Snowflake** using **Azure Data Factory**, while designing an optimized database architecture and creating a solution for efficient data transfer from Azure BLOB storage to SQL Server.
- Design and manage ETL workflows with **SSIS**, ensuring efficient and high-quality data processing and transformation.
- Created and maintained enterprise data pipelines by implementing ETL processes with **MS SQL Server**, **SSIS**, and Azure Data Factory, ensuring smooth data extraction, transformation, and loading.
- Utilized **Azure Synapse Analytics** for data warehousing and advanced querying, enhancing financial data analysis, reporting capabilities, and delivering quicker insights.
- Developed a distributed data processing system with Hadoop and Spark, enabling the analysis of Metabytes of financial data to support advanced analytics and drive innovation in trading strategies.
- Created and deployed a cost-efficient data archiving solution with **Azure Blob Storage**, ensuring long-term data accessibility, regulatory compliance, and reduced storage expenses.
- Collaborated with business users to collect requirements, design visualizations, and deliver training for self-service **BI tools**, improving user experience and accessibility.
- Created migration strategies for transitioning traditional systems to Azure, using lift-and-shift methods and third-party tools to ensure smooth transitions.
- Enhanced existing data movement frameworks by incorporating **automation** and new features to improve reliability and performance.
- **Prioritized automation** to minimize manual intervention and streamline deployment processes across data pipelines.
- Recommended cost-efficient Azure data infrastructure architectures, offering suggestions to optimize spending and resource allocation.
- Applied Python in **CI/CD pipelines** to automate the deployment of database solutions and BI applications on Azure, ensuring continuous integration, testing, and smooth delivery of updates.
- Created and implemented advanced data processing pipelines using **Apache Flink**, enabling complex data transformations and analytics.
- Applied data validation and cleansing techniques to maintain the integrity and accuracy of data stored in Azure environments.
- Worked on code reviews with team members to ensure compliance with OOP principles and best practices, fostering a culture of quality and maintainability.

Environment: Azure Data Factory, Snowflake, PySpark, Power BI, SSIS, Databricks and Apache Flink.

Client: Prisol Soft Technologies, Pvt. Ltd, Hyderabad, India

June 2020 – July 2022

Role: Sr. Data Analyst

Responsibilities:

- Worked on the development of a **Real-Time Inventory and Sales Analytics System** using **Azure Data Factory (ADF)** to improve **inventory management** efficiency and offer valuable insights into sales performance.
- Designed, developed, and maintained reliable data pipelines using **Apache Spark** and **Scala** to efficiently process and analyze data, crucial for transforming and aggregating information from various sources such as point-of-sale (POS) systems and warehouse management systems (WMS).
- Created and managed **ETL processes** with **Apache Spark** and **Hive** to extract, transform, and load data into the Hadoop ecosystem, designed to process both batch and streaming data for near real-time inventory updates as sales occurred.
- **Developed MapReduce (YARN)** jobs for data cleaning, access, and validation, enabling efficient management of large datasets and ensuring data integrity.
- Utilized Google Cloud Platform (GCP) to deploy, manage, and monitor large-scale data processing systems, ensuring optimal availability and performance.

- Created and maintained **ETL processes** using **Talend** and **Bash scripting**, along with big data technologies like Hive, **Impala**, and **Spark** to build data ingestion pipelines that manage real-time updates from multiple sources.
- Utilized containerization technologies like **Docker** and **Kubernetes** to improve application deployment, scalability, and reliability.
- Improved data processing and analytics by using Node.js for server-side scripting and API integrations, in combination with Azure services.
- Transferred data from MS SQL Server and Teradata to HDFS using **SQOOP**.
- Used **Azure Kubernetes Service (AKS)** to deploy and manage containerized applications, enabling the efficient implementation of microservices architectures.
- Created CSV files and JSON scripts to deploy the pipeline in **Azure Data Factory (ADF)** for processing data.
- Delivered continuous support and troubleshooting for ETL processes, swiftly identifying and addressing issues to reduce downtime and ensure smooth data flow.
- Applied **Agile methodology** and Scrum to ensure the on-time delivery of high-quality results.
- Transferred existing self-service and ad-hoc reports to Power BI.
- Focused on data migration using SQL, Azure Storage, and Azure Data Factory. Developed data integration and technical solutions for Azure Data Lake Analytics, **Azure Data Lake Storage**, Azure SQL databases, and Azure SQL Data Warehouse to support **Synapse Analytics** and reporting, improving marketing strategies.

Environment: Apache Spark, Docker, Kubernetes, Azure, GCP, SQL, Power BI, Hadoop.

Client: SBI Life Insurance Co. Ltd., Hyderabad, India

June 2017 – May 2020

Role: Data Analyst

Responsibilities:

- Collaborated on a **Customer Retention and Policy Analytics** design, along with other strategic initiatives, to improve data-driven decision-making and streamline operational processes.
- Worked on analyzing customer behavior, policy renewals, and identifying trends in policy cancellations. This involved utilizing **SQL** and **Python** for data analysis and pattern recognition to pinpoint high-risk customers for churn, while proposing strategies to enhance retention rates.
- Maintained consistent data quality by conducting regular audits, resulting in a 20% decrease in reporting errors.
- Utilized Excel functions for data sorting, analysis, and generating statistical reports.
- Leveraged **SQL** and **Python** to analyze extensive organizational data, uncovering potential cost-saving opportunities.
- Developed several **Directed Acyclic Graphs (DAGs)** to automate ETL pipelines, optimizing data workflows and improving the efficiency of data processing tasks.
- Utilized **Informatica Repository Manager** to manage all repositories (development, test, and validation) and participated in migrating folders between repositories.
- Created a predictive model using machine learning techniques, leading to a 35% improvement in sales forecast accuracy.
- Worked on cross-departmental effort to implement a new data warehousing architecture, resulting in a 25% reduction in data retrieval times.
- Developed new business intelligence reports in **Tableau**, enabling data-driven decision-making.
- Proficient in using tools like **Microsoft Excel** and **Access** to create dashboards and reports.
- Effectively used data validation tools and SQL queries to thoroughly investigate data discrepancies, providing recommendations for process improvements that boost overall data quality.
- Contributed to the analysis, specification, design, implementation, and testing stages of the **Software Development Life Cycle (SDLC)**.

Environment: Excel, SQL, Power BI, Python, Tableau, data cleaning, data warehousing.